

4.2 Lymphoid Tissues

▼ Why does our body need lymphoid tissues?

- Lymphoid tissue is required because **T and B cells cannot express a large enough number of TCRs and BCRs** (T/B Cell Receptors) **to deal with all the possible antigens** that are threatening to the immune system
 - The number of BCRs/TCRs that can be expressed (repertoire) is much larger ($10^{15} - 10^{20}$) than the number of these cells present in our blood ($\sim 7.5 \times 10^9$)
- The lymphoid tissues are introduced to **provide the immune system with an alternative way to combat these antigens and cope with the strenuous demands**
- They **increase the likelihood of interactions between B/T cells and antigens (secondary)**

▼ What comprises the lymphatic system?

- Lymphatic vessels
- Secondary lymphoid tissue

▼ Where does lymphopoiesis occur?

- In the **primary lymphoid tissue** (organs).

▼ What are the three main types of lymphocyte?

- B cells
- T cells
- Natural killer cells.

▼ What is the role of primary lymphoid organs?

- Lymphopoiesis (lymphocyte production)

▼ Recall three primary lymphoid organs.

- Thymus
- Bone marrow
- **Foetal liver**

- ▼ Image showing B and T cell responses to pathogens.

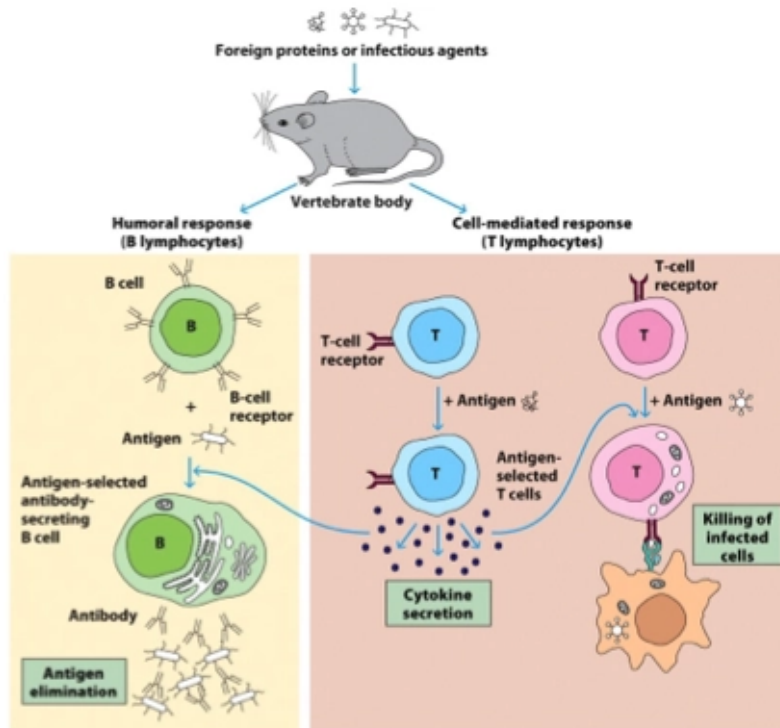


Figure 1-5
Kuby Immunology, Seventh Edition
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- ▼ What is the primary source of haematopoiesis in humans?
- Bone marrow
- ▼ How is haematopoiesis affected during infection?
- **White blood cell (leukocyte) production increases.**
 - T cells and B cells which are lymphocytes are used in the immune response so it makes sense to have more present in response to antigens.
- ▼ Where do B cells undergo their final differentiation?
- In the **bone marrow**.
 - This is why they are called B cells.
- ▼ What is a repertoire in immunological terms?
- The **range of genetically distinct B and T cell receptors present** in a given host
 - The more BCRs or TCRs present in a given host, the more antigens and hence threats can be dealt with
- ▼ Where are antigens found?
- On the **surface of cells, viruses, bacteria etc.**

▼ Where do T cells undergo their final differentiation?

- In the **thymus**
- This is why they are called T cells.

▼ Describe how positive selection in B/T cell differentiation works.

- It is a process that is used to determine **whether a B/T cell can detect antigens using its BCR/TCR (its primary function)**
- If it cannot, it has developed abnormally and **these cells undergo apoptosis.**

▼ Describe how negative selection in B/T cell differentiation works.

- It is a process which is used to determine **whether a B/T cell has BCRs/TCRs that react against self (host cell) antigens**
- We don't want those **B/T cells that react against self antigens**, and so **they undergo apoptosis**

▼ What happens to T cell production as we age, and why?

- The rate of T cell production decreases and this is due to a process called **thymus involution**
- This involves the **thymus shrinking**, and being **reduced to fat** and so the **production of T cells** is reduced because the thymus is involved in T cell differentiation.

▼ What are secondary lymphoid organs?

- Organs which act as the site where **lymphocytes can interact with antigens and other lymphocytes.**

▼ Approximately what percentage of the total number of T cells are contained in the blood?

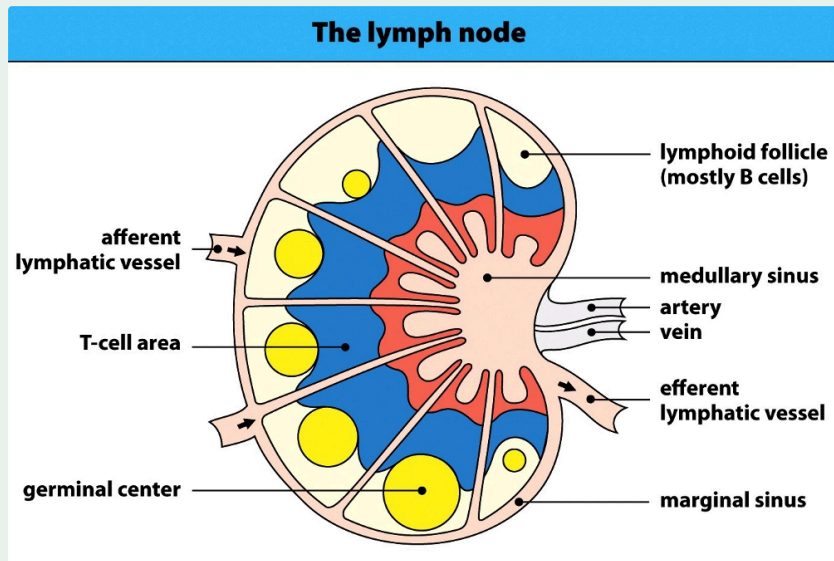
- 2%
- The rest is found mainly in the secondary lymphoid organs

▼ Recall four secondary lymphoid organs.

- Spleen
- Lymph nodes
- Appendix
- Tonsils

▼ What are afferent and efferent lymphatic vessels?

- An **afferent vessel** is where lymph fluid flows **into the lymph node**
- An **efferent vessel** is where lymph fluid flows **out of the lymph node**



▼ What does the lymph fluid contain?

- Lymphocytes
- Antigens
- Although the lymph nodes have arterial and venous connections, these two elements flow to the lymph nodes via the lymphatic vessels.

▼ What causes the separation between the B cells and T cells within lymph nodes?

- **Chemokines**
- They are **produced by the stroma within the lymph nodes**

▼ What additional role does the spleen play that the lymph nodes don't play?

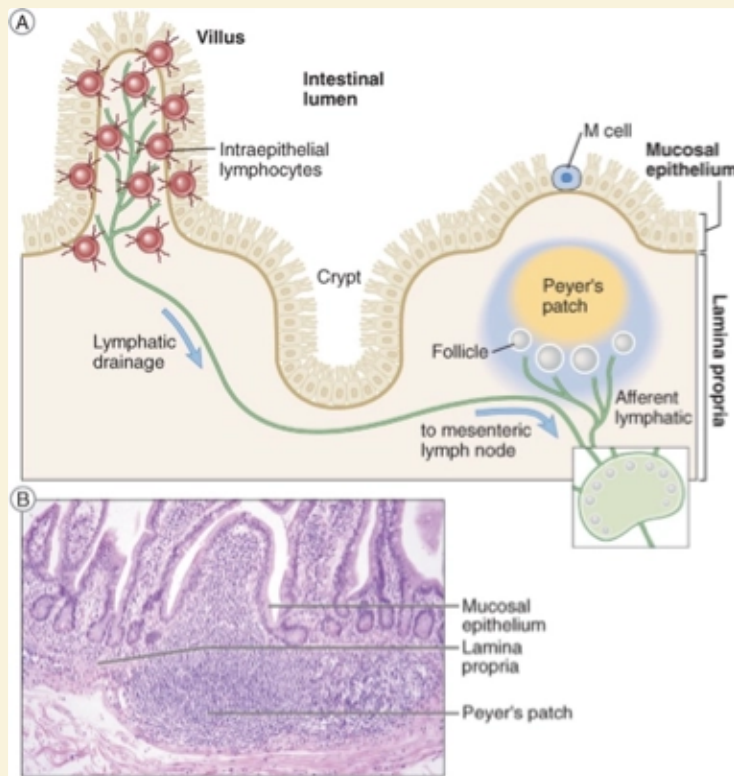
- The spleen filters **blood antigens** when it comes through the arteries
- The lymph nodes filter **skin or respiratory tract antigens**

▼ What is the first line of defence against infection?

- Mucosal/epithelial barriers

▼ What are Peyer's patches?

- These are specialised secondary lymphoid tissues that are found below the epithelium in the **ileum of the small intestine**



▼ Describe a property of Peyer's patches.

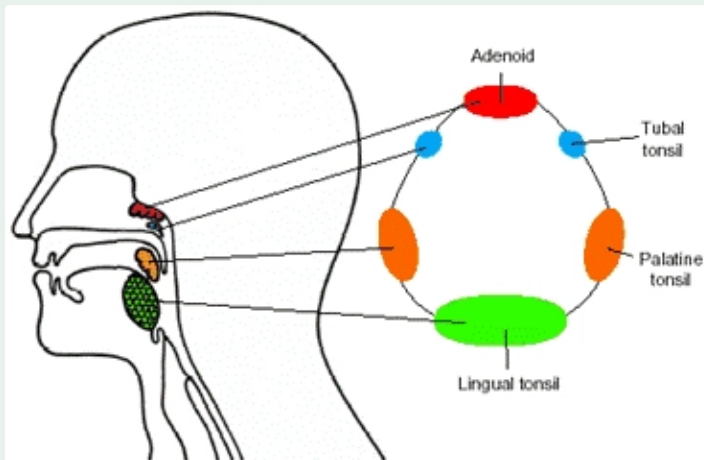
- They have a **high frequency of B cells**
- They have a **lot of germinal centres**

▼ What is a germinal centre?

- A site where **B cells undergo mutation and selection to produce high-affinity antibodies**

▼ What is Waldeyer's ring?

- The ring of **tonsils** that **surround the pharyngeal region**

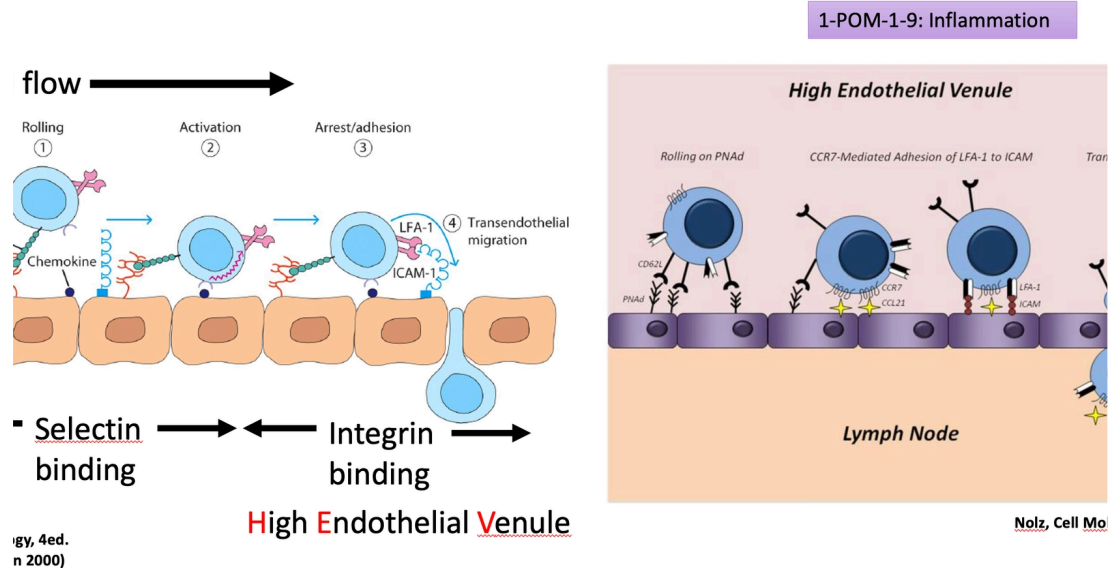


▼ Why is it a better idea for B and T cells to be located within secondary lymphoid tissue, rather than freely circulating around the blood?

- A lot of **fluid that surrounds tissues is drained through the secondary lymphoid organs** for filtering and is returned to the blood
 - 2-3 L of fluid returned to blood via superior vena cava per day
- If there are T and B cells located within these tissues, then there is a **far greater chance** of them **encountering antigens** that flow through these systems and dealing with them
- Lymphocytes also recirculate around the blood once every 24 hours and this is another way of **increasing the chance of an antigen-lymphocyte interaction**.
- Overall, the **purpose of the secondary lymphoid organs** is to **increase the speed at which the lymphocytes travel across the body**, so that they can **deal with their respective antigens swiftly**.
- This is needed because our **T Cell repertoire (or number of different antigens) vastly outweighs the number of T cells we actually have**.

- ▼ Image showing how chemokines, integrins and selectins are involved in the extravasation of T cells into the lymph node.
 - **CCR7** binds to its chemokine **CCL21**
 - **LFA-1** binds to its chemokine **ICAM**

Extravasation of naïve T cells into lymph nodes



- ▼ How does inflammation affect T cell and B cell distribution?
 - Inflammation at a particular site will cause signals to be released at this site which will instruct the specific T cells and B cells at the site of inflammation to remain where they are, to increase the chance of them encountering their respective antigens on an antigen presenting cell.
 - Normally, T cells and B cells are circulating through the lymphatic system, but inflammation sends signals for them to migrate to the site of inflammation
- ▼ What is an antigen-presenting cell (APC)?
 - An APC is a cell type which **presents antigens captured from the environment**, so that a T cell can recognise it and initiate an adaptive immune response.